

Title: Ethical Reflection on the Use of the COMPAS Recidivism Dataset in Predictive Modeling

Introduction

The COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) recidivism dataset has become a prominent example in the discourse on algorithmic fairness and ethical concerns in criminal justice applications. In a recent data science project involving this dataset, I explored the use of predictive modeling to assess the likelihood of recidivism. Recognizing the ethical implications of such work, I ensured that the project adhered to established ethical AI principles.

Transparency and Interpretability

To maintain transparency, I opted for interpretable models such as logistic regression and decision trees rather than complex black-box algorithms. This choice allowed for greater clarity in understanding how each variable contributed to the prediction outcomes. Stakeholders—including non-technical users—were able to review the decision-making process, which is essential in sensitive applications like criminal justice.

Fairness and Bias Mitigation

One of the primary ethical challenges with the COMPAS dataset is the potential for racial bias. Previous studies have shown disparities in false positive rates across racial groups. To address this, I conducted a fairness audit by analyzing model performance metrics (e.g., precision, recall, false positive rate) disaggregated by race. I applied fairness-aware machine learning techniques such as reweighting and adversarial debiasing to reduce these disparities and improve fairness in model outputs.

Data Ethics and Accountability

The dataset itself is derived from a proprietary system whose scoring mechanism is not fully transparent. Treating COMPAS risk scores as ground truth without critique would be ethically flawed. I documented these limitations clearly and discouraged the blind adoption of these scores in my project. Moreover, I emphasized that any model trained using this dataset should only serve as a decision-support tool, not a decision-maker.

Social Impact and Responsible AI Use

Understanding the social implications of automated decision-making, especially in the criminal justice system, was central to my project design. I advocated for human-in-the-loop systems where final decisions are made by qualified professionals supported by, but not dependent on, algorithmic recommendations. Involving policymakers, ethicists, and community representatives in reviewing the model was a key recommendation.

Conclusion

This project reinforced the importance of embedding ethical principles in every phase of AI development. Beyond technical accuracy, responsible AI requires deliberate actions to ensure fairness, accountability, and respect for human rights. My approach to the COMPAS dataset aimed to reflect these values while contributing to a more equitable use of predictive analytics in sensitive domains.